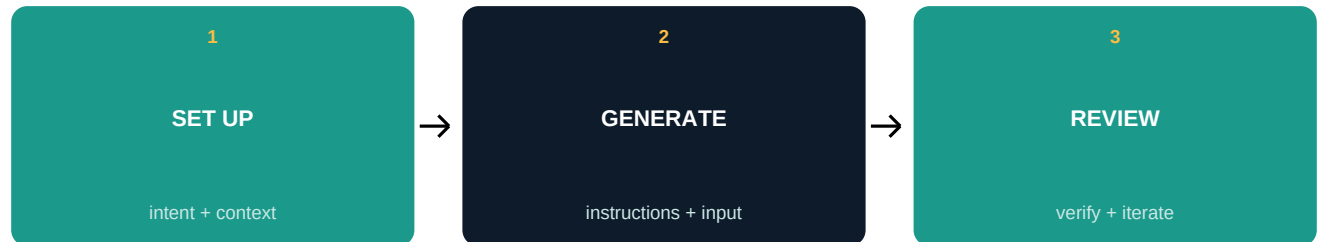


CHAPTER 2

The AI Interaction Stack, What Actually Determines Output Quality



INTRODUCING A MAP When a piece of work goes wrong, it helps to know where in the process the problem occurred. A cake that doesn't rise failed at a different point than a cake that doesn't taste right. The diagnosis matters.

AI interactions fail at predictable places. This chapter introduces a model, the AI Interaction Stack, that maps those places. It gives you a vocabulary for diagnosing what went wrong and a checklist for getting it right before you start.

THE AI INTERACTION STACK The AI Interaction Stack has eleven layers. Each layer represents something that shapes what AI produces. When an interaction produces poor results, at least one of these layers is inadequate or missing.

You do not need all eleven layers for every task. A quick factual question might only need layers 1-4. A complex document workflow might need all eleven. The point is knowing which layers exist so you can deliberately choose which ones to activate.

Here they are, in order:

Read the stack from the bottom up. Start with the real goal, then add only the layers that this task needs. The upper layers help you check, improve, and continue the work; they do not replace the clarity of the layers below them.

THE AI INTERACTION STACK Read from the bottom up. Activate only the layers this task needs.

11 Iteration What happens next?

10 Verification How will you check it?

9 Output What should the result look like?

8 Workflow One step or several?

7 Tools What capabilities are available?

6 Workspace What context can persist?

5 Source material What can AI work from?

4 Examples What does good look like?

3 Instructions What exactly should AI do?

2 Context What does AI need to know?

1 Intent What are you trying to achieve?

LAYER 1, INTENT What are you actually trying to achieve? Not the surface task ("write a summary") but the underlying goal ("help my CEO decide quickly"). These are often different, and the difference matters.

A summary intended to help someone decide looks different from a summary intended to brief someone who already decided. Same task, different intent, different output.

Before you start: Can you state your actual goal in one sentence?

LAYER 2, CONTEXT What does AI need to know about your situation? Who is involved? What is the background? What constraints apply? What has already been tried?

Context is the layer most people consistently underinvest in. It is also the layer where providing more almost always pays off.

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Before you start: What does AI not know that would change its answer?

LAYER 3, INSTRUCTIONS What exactly should AI do? This is distinct from intent. Intent is the goal; instructions are the specific action you want AI to take to achieve it.

"Summarise this document" is an instruction. "Summarise this document in three bullet points, prioritising financial implications, written for a non-specialist reader" is a better one.

Before you start: How specific are your instructions? Could a capable human follow them without asking a question?

LAYER 4, EXAMPLES What does good look like? Sometimes the most efficient way to specify a tone, structure, or style is to show AI an example of it, a previous email you wrote, a format you liked, a sample of the style you want.

Examples do more than description. Telling AI to be "warm but professional" is imprecise. Showing it a warm-but-professional email is not.

Before you start: Is there an example that would make the target output clearer?

LAYER 5, SOURCE MATERIAL What documents, data, notes, or materials should AI work from? AI can read files, contracts, reports, research papers, meeting notes, spreadsheets, and its output will be grounded in what you provide rather than in generalisation.

When you ask AI to help with a specific document or dataset, provide it. Don't describe it, provide it.

Before you start: What source materials could improve the specificity and accuracy of the output?



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LAYER 6, WORKSPACE Have you set up a persistent context, a project, a set of standing instructions, a shared background, so you don't have to repeat essential information every time?

For tasks you do repeatedly, a workspace eliminates repetition and builds consistency. Part III is dedicated to this layer.

Before you start: Is this a recurring task where a workspace would save significant setup time?

LAYER 7, TOOLS What capabilities does AI have access to in this context? Can it search the web? Run calculations? Read uploaded files? Generate images? Access calendar or email?

Knowing which tools are available, and which aren't, shapes what you can reasonably ask for and where you need to verify externally.

Before you start: What tools are available? Does this task benefit from any of them?

LAYER 8, WORKFLOW Is this a single-step task or a multi-step process? Complex work almost always benefits from decomposition: breaking the task into stages, running each stage separately, reviewing and adjusting between stages.

Single-prompt attempts at complex tasks produce mediocre results that are hard to improve. Staged workflows produce better outputs at each stage and make errors easier to catch.

Before you start: Is this a single-step task, or should it be broken into stages?

LAYER 9, OUTPUT What should the result look like? Format, length, structure, tone, audience, medium, all of these shape the output and all are specifiable.

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Without output specifications, AI defaults to whatever format its training suggests is most common for this type of request. That default is often wrong for your specific situation.

Before you start: Have you specified format, length, tone, and structure?

LAYER 10, VERIFICATION How will you check the output? What specifically might be wrong? AI makes errors, factual, logical, and interpretive, with calm confidence. Your verification approach should match the stakes of the task and the risk profile of the content.

Telling AI to flag its own uncertainties is a useful first step. It doesn't replace your review, but it helps.

Before you start: How will you verify this output before you use or share it?

LAYER 11, ITERATION How will you use this output as the starting point for the next stage? AI interactions are rarely single exchanges, they are conversations where each response informs the next prompt. Good output at layer 11 becomes input at layer 2 of the next pass.

Before you start: Is this a one-shot task, or will you need to iterate? What will the next step be?

WHY IT WORKS THE STACK IS NOT A CHECKLIST TO COMPLETE IN FULL BEFORE EVERY INTERACTION. IT IS A DIAGNOSTIC F



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THE STACK IN PRACTICE Here is the same task at two different levels of stack activation:

LOW STACK (LAYERS 1, 3 ONLY): "Summarise this meeting."

Result: A generic summary with equal weight given to all topics discussed. Possibly bulleted, possibly in paragraphs. Pitched at a generic reader. Takes five minutes to edit into something useful.

HIGH STACK (LAYERS 1-5, 9-10): "I need a meeting summary for three people who weren't present: my manager, the client's account manager, and one of our developers. Each person needs to see different things. My manager wants the key decisions and next steps only. The client account manager wants what the client committed to. The developer wants the technical requirements we discussed. The attached transcript is the source. Write three separate summaries, one for each person, no longer than half a page each. Flag anything that seemed unresolved or ambiguous."

Result: Three targeted summaries, immediately usable, with potential problems flagged.

The difference is not length of prompt. It is activation of the right layers for the task.

TRY THIS THINK OF AN AI TASK YOU NEED TO DO THIS WEEK. WRITE DOWN WHAT LAYERS OF THE STACK YOU WOULD NORM

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WHICH LAYERS MATTER MOST? The research on what makes AI interactions succeed points consistently to three layers as the highest-leverage investments for most people in most tasks:

Context (Layer 2) is the most consistently underinvested layer and the one with the highest return. Providing relevant context costs little and changes output significantly.

Instructions (Layer 3) are the most frequently underdeveloped layer. Most instructions are too vague. The skill of writing specific, actionable instructions is developed in Chapter 7.

Verification (Layer 10) is the most commonly skipped layer. Part VI covers it in depth. Skipping it is also the source of most AI-related professional embarrassment.

WATCH OUT THE STACK IS A MENTAL MODEL, NOT A PROCEDURE. IT DOES NOT MEAN YOU SHOULD SPEND TWENTY MINUTES



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